DEFINITION A cipher provides ***misuse-resistant authenticated encryption (MRAE)*** if accidental or deliberate nonce reuse results in only a small loss of security. An attacker can only learn if the same message has been encrypted twice with the same nonce and key and there is no loss of authenticity. ***Synthetic Initialization Vector (SIV) mode*** is a well-known MRAE mode, and ***AES-SIV*** the most common use of it.

SIV mode works by computing the nonce (also known as an Initialization Vector or IV) using a pseudorandom function (PRF) rather than using a purely random value or counter. Many MACs used for authentication are also PRFs, so SIV reuses the MAC used for authentication to also provide the IV, as shown in figure 12.8.

CAUTION Not all MACs are PRFs so you should stick to standard implementations of SIV mode rather than inventing your own.

